INFO8006 Introduction to Artificial Intelligence

Exercise session 5: Reasoning over time

Recap

Markov models

The world is a discrete series of time slices, where $\mathbf{X}_t$ denotes the set of *unobservable* state variables and $\mathbf{E}_t$ the set of *observable* evidence variables at time t. We specify a prior $\mathbf{P}(\mathbf{X}_0)$, a *transition model* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1})$ and a *sensor model* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1})$, under

- the first-order *Markov assumption* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1}) = \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{t-1})$, which defines a first-order *Markov process*;
- the *sensor Markov assumption* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1}) = \mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$;
- the *stationarity assumption*: the transition and sensor models are the same for all t.

The joint distribution of all variables up to t is then

$$\mathbf{P}(\mathbf{X}_{0:t}, \mathbf{E}_{1:t}) = \mathbf{P}(\mathbf{X}_0) \prod_{i=1}^{t} \mathbf{P}(\mathbf{X}_i \mid \mathbf{X}_{i-1}) \mathbf{P}(\mathbf{E}_i \mid \mathbf{X}_i).$$

Inference tasks

Given the evidence $\mathbf{e}_{1:t}$ collected up to time t, we define

- *prediction*: $\mathbf{P}(\mathbf{X}_{t+k} \mid \mathbf{e}_{1:t})$ for $k > 0$;
- *filtering*: $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$, the *belief state* about the current hidden state $\mathbf{X}_t$;
- *smoothing*: $\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t})$ for $0 \leq k < t$;
- *most likely explanation*: $\arg\max_{\mathbf{x}_{1:t}} P(\mathbf{x}_{1:t} \mid \mathbf{e}_{1:t})$.

Prediction

To predict the future, the belief state is pushed through the transition model,

$$\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}) = \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}),$$

and this is repeated up to $t + k$.

Filtering

The *Bayes filter* maintains the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ as time passes and new evidence $\mathbf{e}_{t+1}$ is collected. It alternates between a predict step, which projects the current belief state forward from t to $t + 1$ through the transition model, and an update step, which updates this new state using the evidence $\mathbf{e}_{t+1}$:

$$\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}).$$

The belief state is a *forward message* $\mathbf{f}_{1:t}$, with $\mathbf{f}_{1:t+1} \propto \text{forward}(\mathbf{f}_{1:t}, \mathbf{e}_{t+1})$. For continuous variables, sums are replaced with integrals and probability mass functions with densities:

$$p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto p(\mathbf{e}_{t+1} \mid \mathbf{x}_{t+1}) \int p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) p(\mathbf{x}_t \mid \mathbf{e}_{1:t}) \, \mathrm{d}\mathbf{x}_t.$$

Smoothing

Dividing the evidence $\mathbf{e}_{1:t}$ into $\mathbf{e}_{1:k}$ and $\mathbf{e}_{k+1:t}$,

$$\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t}) \propto \mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k) = \alpha \, \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t},$$

where $\times$ is a pointwise multiplication of vectors and the *backward message* $\mathbf{b}_{k+1:t} = \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k)$ is computed by backwards recursion, $\mathbf{b}_{k+1:t} = \text{backward}(\mathbf{b}_{k+2:t}, \mathbf{e}_{k+1})$.

Most likely explanation

The *Viterbi algorithm* replaces the forward message with $\mathbf{m}_{1:t} = \max_{\mathbf{x}_{1:t-1}} \mathbf{P}(\mathbf{x}_{1:t-1}, \mathbf{X}_t \mid \mathbf{e}_{1:t})$, the probability of the most likely path to each state, computed as

$$\mathbf{m}_{1:t+1} \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \max_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) \mathbf{m}_{1:t}.$$

Hidden Markov models

A *hidden Markov model* (HMM) is a Markov process in which the state $\mathbf{X}_t = X_t$ and the evidence $\mathbf{E}_t = E_t$ are both single discrete random variables, with domains $D_{X_t} = \{1, \ldots, S\}$ and $D_{E_t} = \{1, \ldots, R\}$. The forward-backward algorithm then reduces to matrix-vector operations:

- the prior $\mathbf{P}(X_0)$ becomes a column vector $\mathbf{f}_0 \in \mathbb{R}_+^S$;
- the transition model $\mathbf{P}(X_t \mid X_{t-1})$ becomes an $S \times S$ *transition matrix* $\mathbf{T}$ such that $\mathbf{T}_{ij} = P(X_t = j \mid X_{t-1} = i)$;
- the sensor model $\mathbf{P}(E_t \mid X_t)$ is an $S \times R$ *sensor matrix* $\mathbf{B}$ such that $\mathbf{B}_{ij} = P(E_t = j \mid X_t = i)$;
- the *observation matrix* $\mathbf{O}_t$ is a diagonal matrix whose elements correspond to the column e_t of $\mathbf{B}$.

With column vectors for the forward and backward messages,

$$\mathbf{f}_{1:t+1} = \alpha \mathbf{O}_{t+1} \mathbf{T}^T \mathbf{f}_{1:t},$$
$$\mathbf{b}_{k+1:t} = \mathbf{T} \mathbf{O}_{k+1} \mathbf{b}_{k+2:t},$$

where $\mathbf{b}_{t+1:t}$ is an all-one vector of size S.

Stationary distribution (optional)

Without new evidence, the predicted distribution may converge, as k grows, to a *stationary distribution*: a fixed point $\mathbf{P}(\mathbf{X}_{t+1}) = \mathbf{P}(\mathbf{X}_t)$ of the predict step $\mathbf{P}(\mathbf{X}_{t+1}) = \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t)$. For an HMM, it is a vector $\mathbf{f}$ such that $\mathbf{f} = \mathbf{T}^T \mathbf{f}$.

Matrix multiplication

In index notation,

$$\mathbf{C} = \mathbf{A}\mathbf{B} \quad \Leftrightarrow \quad \mathbf{C}_{ij} = \sum_u \mathbf{A}_{iu} \mathbf{B}_{uj} \quad \forall \, i, j,$$
$$\mathbf{d} = \mathbf{A}\mathbf{f} \quad \Leftrightarrow \quad \mathbf{d}_i = \sum_u \mathbf{A}_{iu} \mathbf{f}_u \quad \forall \, i.$$

Cheat sheet for Gaussian distributions

If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix}, \begin{pmatrix}\mathbf{A} & \mathbf{C}\\ \mathbf{C}^T & \mathbf{B}\end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a}, \mathbf{A}),\\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b}, \mathbf{B}),\\ p(\mathbf{x} \mid \mathbf{y}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a} + \mathbf{C}\mathbf{B}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{C}\mathbf{B}^{-1}\mathbf{C}^T),\\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

If $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \mathbf{m}, \mathbf{P})$ and $p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(\mathbf{y} \mid \mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R})$, then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{m}\\ \mathbf{H}\mathbf{m} + \mathbf{u}\end{pmatrix}, \begin{pmatrix}\mathbf{P} & \mathbf{P}\mathbf{H}^T\\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R}\end{pmatrix}\right).$$

Kalman filter

The *Kalman filter* is a special case of the Bayes filter for continuous state and evidence variables, which assumes a Gaussian prior $p(\mathbf{x}_0)$, a linear Gaussian transition model and a linear Gaussian sensor model,

$$\begin{aligned} p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t+1} \mid \mathbf{A}\mathbf{x}_t + \mathbf{b}, \mathbf{Q}),\\ p(\mathbf{e}_t \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{e}_t \mid \mathbf{H}\mathbf{x}_t + \mathbf{d}, \mathbf{R}). \end{aligned}$$

Under these assumptions, the predict and update steps both yield Gaussian distributions, $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}^-_{t+1}, \boldsymbol{\Sigma}^-_{t+1})$ and $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}_{t+1}, \boldsymbol{\Sigma}_{t+1})$, whose parameters are derived in closed form with the Gaussian identities above.

Exercises

In session: exercises 1, 2.

Exercise 1 Load of a shared server (Exam, January 2023)

A scheduling program submits jobs to a shared compute server, whose load varies over time. Each job runs under either a heavy or a light load. From past logs, the program knows that if a job runs under a heavy load, there is a 40% chance that the next job also runs under a heavy load. If a job runs under a light load, there is a 70% chance that the next job also runs under a light load. The program cannot see the load, but it sees the response time of each job, which is either fast, normal or slow. If the load is heavy, the response time is fast with probability 20% and slow with probability 60%. If the load is light, it is fast with probability 50% and slow with probability 30%.

(a) Define the components of a hidden Markov model (HMM), where the state $X_t \in \{\texttt{heavy}, \texttt{light}\}$ is the load under which the t-th job runs and the evidence $E_t \in \{\texttt{fast}, \texttt{normal}, \texttt{slow}\}$ is its response time.

Solution.

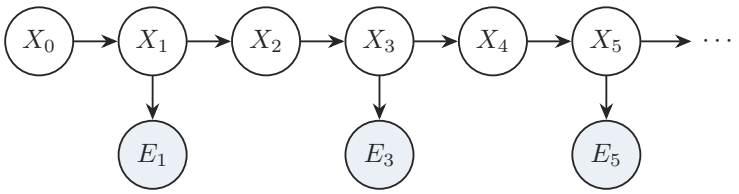
$$f_0 = P(X_0) = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

$$T = P(X_t \mid X_{t-1}) = \begin{pmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{pmatrix}$$

$$B = P(E_t \mid X_t) = \begin{pmatrix} 0.2 & 0.2 & 0.6 \\ 0.5 & 0.2 & 0.3 \end{pmatrix}$$

(b) At some point, the server stops reporting the response time of every other job (the 2nd, the 4th, ...), so that the program only observes the 1st, 3rd, 5th, ... jobs. Sketch a Bayesian network describing this new process.

Solution.



(c) Express, in terms of the HMM components, the distribution $P(X_t \mid e_{1:t:2})$ of the load for the t-th job given the observations $e_{1:t:2}$ of every other job. Note that, $e_{1:6:2} = (e_1, e_3, e_5)$ and $e_{1:7:2} = (e_1, e_3, e_5, e_7)$. Separate the cases where t is even and odd.

Solution. Denoting $f_t = P(X_t \mid e_{1:t:2})$,

$$f_{2n} = T^T f_{2n-1}$$
$$f_{2n+1} = \alpha O_{2n+1} T^T f_{2n}$$

The two formulae above can be readily derived from an example. For $2n = 4$:

$$\begin{aligned}
\forall x_4 : P(X_4 = x_4 \mid e_{1,3}) &= \sum_{x_3} P(X_4 = x_4 \mid X_3 = x_3, e_{1,3}) P(X_3 = x_3 \mid e_{1,3}) \\
&= \sum_{x_3} P(X_4 = x_4 \mid X_3 = x_3) P(X_3 = x_3 \mid e_{1,3}) \\
&\quad \text{(Under the Markov assumption)} \\
\Leftrightarrow \forall j : f_{4,j} &= \sum_u T_{u,j} f_{3,u} \\
&= \sum_u (T^T)_{j,u} f_{3,u} \\
\Leftrightarrow f_4 &= T^T f_3
\end{aligned}$$

For $2n + 1 = 3$:

$$\begin{aligned}
\forall x_3 : P(X_3 = x_3 \mid e_{1,3}) &= \alpha P(e_3 \mid X_3 = x_3, e_1) P(X_3 = x_3 \mid e_1) \\
&\quad \text{(As } P(A \mid B, C) P(B \mid C) = P(B \mid A, C) P(A \mid C)\text{)} \\
&= \alpha P(e_3 \mid X_3 = x_3) P(X_3 = x_3 \mid e_1) \\
&\quad \text{(Under the Markov assumption)} \\
&= \alpha P(e_3 \mid X_3 = x_3) \sum_{x_2} P(X_3 = x_3 \mid X_2 = x_2, e_1) P(X_2 = x_2 \mid e_1) \\
&= \alpha P(e_3 \mid X_3 = x_3) \sum_{x_2} P(X_3 = x_3 \mid X_2 = x_2) P(X_2 = x_2 \mid e_1) \\
&\quad \text{(Markov assumption)} \\
\Leftrightarrow \forall i : f_{3,i} &= \alpha B_{i,e_3} \sum_u T_{u,i} f_{2,u} \\
&= \alpha B_{i,e_3} \sum_u (T^T)_{i,u} f_{2,u} \\
&= \alpha \sum_j O_{3,i,j} \sum_u (T^T)_{j,u} f_{2,u} \\
&\quad \text{(As } O_3 \text{ is a diagonal matrix)} \\
\Leftrightarrow f_3 &= \alpha O_3 T^T f_2
\end{aligned}$$

(d) Given the observations $e_{1:3:2} = (e_1, e_3) = (\texttt{slow}, \texttt{fast})$, compute the probability that the 3rd job runs under a heavy load.

Solution.

$$f_1 = \alpha_1 O_1 T^T f_0 = \alpha_1 \begin{pmatrix} 0.6 & 0 \\ 0 & 0.3 \end{pmatrix} \begin{pmatrix} 0.4 & 0.3 \\ 0.6 & 0.7 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} \approx \begin{pmatrix} 0.5185 \\ 0.4815 \end{pmatrix}$$

$$f_2 = T^T f_1 = \begin{pmatrix} 0.4 & 0.3 \\ 0.6 & 0.7 \end{pmatrix} \begin{pmatrix} 0.5185 \\ 0.4815 \end{pmatrix} \approx \begin{pmatrix} 0.3519 \\ 0.6481 \end{pmatrix}$$

$$f_3 = \alpha_3 O_3 T^T f_2 = \alpha_3 \begin{pmatrix} 0.2 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} 0.4 & 0.3 \\ 0.6 & 0.7 \end{pmatrix} \begin{pmatrix} 0.3519 \\ 0.6481 \end{pmatrix} \approx \begin{pmatrix} 0.1678 \\ 0.8322 \end{pmatrix}$$

Finally, $P(X_3 = \texttt{heavy} \mid e_{1:3:2}) \approx 0.1678$.

Exercise 2 Umbrella world (AIMA ch. 14)

A security robot guards a secret underground installation. It would like to know whether it is raining today, but its only access to the outside world occurs each morning, when it sees the director coming in with, or without, an umbrella. For each day t, the evidence is a single variable $Umbrella_t \in \{1, 0\}$, *i.e.* whether the umbrella appears or not, and the (hidden) state is a single variable $Rain_t \in \{1, 0\}$, *i.e.* whether it is raining or not.

The robot's model assumes that from one day $t-1$ to the next t, the chances that the weather stays the same are 70 %. It also assumes that the director brings an umbrella 90 % of the time when it is raining, and 20 % of the time otherwise.

(a) The robot represents the umbrella world as a Markov model. What formal assumptions correspond to these beliefs?

Solution. The first belief describes the umbrella world as a *first-order Markov process*, such that

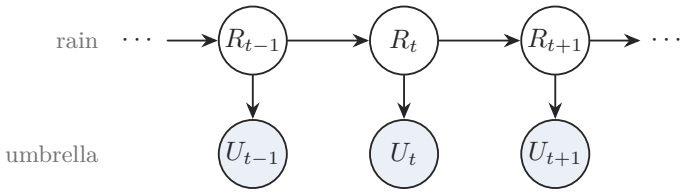
$$P(R_t \mid R_{1:t-1}) = P(R_t \mid R_{t-1}),$$

i.e. R_t is independent from $R_{1:t-2}$ given R_{t-1}. The second belief is equivalent to a *first-order sensor Markov assumption*, implying that

$$P(U_t \mid R_{1:t}, U_{1:t-1}) = P(U_t \mid R_t).$$

(b) Sketch a Bayesian network structure describing the umbrella world and provide the transition and sensor models.

Solution.



R_{t-1}	$P(R_t = 1 \mid R_{t-1})$
1	0.7
0	0.3

R_t	$P(U_t = 1 \mid R_t)$
1	0.9
0	0.2

This corresponds to the following transition and sensor matrices

$$T = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} \quad B = \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix}.$$

(c) Express the distributions $P(R_{t+1} \mid R_{t-1})$, $P(U_t \mid R_{t-1})$ and $P(R_t \mid R_{t-1}, U_t)$ in terms of the transition and sensor models.

Solution.

$$\begin{aligned}
P(R_{t+1} \mid R_{t-1}) &= \sum_{r_t} P(R_{t+1} \mid r_t) P(r_t \mid R_{t-1}) \\
&= TT \\
&= \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} = \begin{pmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{pmatrix} \\
P(U_t \mid R_{t-1}) &= \sum_{r_t} P(U_t \mid R_{t-1}, r_t) P(r_t \mid R_{t-1}) \\
&= \sum_{r_t} P(U_t \mid r_t) P(r_t \mid R_{t-1}) \\
&= TB \\
&= \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix} = \begin{pmatrix} 0.69 & 0.31 \\ 0.41 & 0.59 \end{pmatrix} \\
P(R_t \mid R_{t-1}, U_t) &= \frac{P(U_t \mid R_{t-1}, R_t) P(R_t \mid R_{t-1})}{P(U_t \mid R_{t-1})} \\
&= \frac{P(U_t \mid R_t) P(R_t \mid R_{t-1})}{P(U_t \mid R_{t-1})}
\end{aligned}$$

(d) Suppose the robot observes an unending sequence of days on which the umbrella appears. Show that, as the days go by, the probability of rain on the current day tends monotonically towards a fixed point. Calculate this fixed point.

Solution. We are asked to prove that $x_t = P(R_t = 1 \mid U_{1:t} = 1)$ tends monotonically (with respect to t) towards a fixed point x^*. To do so, we need to find the value of x_t, which corresponds to apply *filtering* on the Markov model defined by the umbrella world. On this basis, we have

$$\begin{aligned}
P(R_t \mid U_{1:t} = 1) &= \alpha P(U_t = 1 \mid R_t) \sum_{r_{t-1}} P(R_t \mid r_{t-1}) P(r_{t-1} \mid U_{1:t-1} = 1) \\
&= \alpha \begin{pmatrix} 0.9 & 0 \\ 0 & 0.2 \end{pmatrix} \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} P(R_{t-1} \mid U_{1:t-1} = 1) \\
\Leftrightarrow \qquad \begin{pmatrix} x_t \\ 1 - x_t \end{pmatrix} &= \alpha \begin{pmatrix} 0.63 & 0.27 \\ 0.06 & 0.14 \end{pmatrix} \begin{pmatrix} x_{t-1} \\ 1 - x_{t-1} \end{pmatrix} \\
&= \alpha \begin{pmatrix} 0.27 + (0.63 - 0.27)\, x_{t-1} \\ 0.14 + (0.06 - 0.14)\, x_{t-1} \end{pmatrix} \\
&= \frac{1}{0.41 + 0.28\, x_{t-1}} \begin{pmatrix} 0.27 + 0.36\, x_{t-1} \\ 0.14 - 0.08\, x_{t-1} \end{pmatrix}.
\end{aligned}$$

Then, if there is a fixed point $x^* \in [0, 1]$, it satisfies

$$\begin{aligned}
x^* &= \frac{0.27 + 0.36\, x^*}{0.41 + 0.28\, x^*} \\
\Leftrightarrow \quad 0.41\, x^* + 0.28\, x^{*2} &= 0.27 + 0.36\, x^* \\
\Leftrightarrow \quad 0 &= 0.28\, x^{*2} + 0.05\, x^* - 0.27 \\
\Rightarrow \quad x^* &= \frac{-0.05 + \sqrt{0.05^2 + 4 \times 0.27 \times 0.28}}{2 \times 0.28} \approx 0.897.
\end{aligned}$$

To show that x_t tends monotonically towards x^*, it is sufficient to prove that x_t is always between x^* and x_{t-1}, *i.e.*

$$(x^* - x_t)(x_t - x_{t-1}) > 0,$$

when $x_{t-1} \neq x^*$, which is left as an exercise to the motivated reader.

(e) Now consider forecasting further and further into the future, given t umbrella observations. Is there a fixed point? If yes, compute its exact value.

Solution. We are asked to forecast the rain probability $x_k = P(R_{t+k} = 1 \mid U_{1:t} = 1)$, *i.e.* to perform *prediction*. We have

$$\begin{aligned} P(R_{t+k} \mid U_{1:t} = 1) &= \sum_{r_{t+k-1}} P(R_{t+k} \mid r_{t+k-1}) P(r_{t+k-1} \mid U_{1:t} = 1) \\ &= \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} P(R_{t+k-1} \mid U_{1:t} = 1) \\ \Leftrightarrow \qquad \begin{pmatrix} x_k \\ 1 - x_k \end{pmatrix} &= \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} x_{k-1} \\ 1 - x_{k-1} \end{pmatrix}. \end{aligned}$$

By symmetry, we find the fixed point $x^* = 0.5$, which highlights the fact that, without new evidences, uncertainty about the states accumulates.

Exercise 3 Coin-tossing machine

The surveillance program of a casino watches a coin-tossing machine. The machine holds 3 biased coins a, b and c, whose probabilities of heads are 80 %, 50 % and 20 %, respectively. It first picks one of the coins at random. Then, 4 times, it either keeps the current coin with probability $\frac{2}{3}$ or replaces it by one of the two other coins, chosen at random, then tosses the coin and shows the result. The program only sees the results: heads, heads, tails, heads.

(a) Provide a hidden Markov model (HMM) that describes the process.

Solution.

- The hidden state at time t is $X_t \in \{a, b, c\}$ and represents the tossed coin.
- The evidence at time t is $E_t \in \{0, 1\}$ and represents the result (heads or tail) of the toss. We have $e_1 = e_2 = e_4 = 0$ and $e_3 = 1$.
- The prior vector
$$f_0 = P(X_0) = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}^T$$
- The transition matrix
$$T = P(X_t \mid X_{t-1}) = \begin{pmatrix} \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{pmatrix}$$
- The sensor matrix
$$B = P(E_t \mid X_t) = \begin{pmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \\ 0.2 & 0.8 \end{pmatrix}$$

(b) What are the probabilities of the last coin given the sequence of results?

Solution. We are asked to calculate the distribution $f_t = P(X_t \mid e_{1:t})$ for $t = 4$. Applying Bayes filter, we have

$$\begin{aligned} P(X_t \mid e_{1:t}) &= \alpha P(e_t \mid X_t, e_{1:t-1}) P(X_t \mid e_{1:t-1}) \\ &= \alpha P(e_t \mid X_t, e_{1:t-1}) \sum_{x_{t-1}} P(X_t \mid x_{t-1}, e_{1:t-1}) P(x_{t-1} \mid e_{1:t-1}) \\ &= \alpha P(e_t \mid X_t) \sum_{x_{t-1}} P(X_t \mid x_{t-1}) P(x_{t-1} \mid e_{1:t-1}) \\ \Leftrightarrow \qquad f_t &= \alpha\, O_t\, T^T\, f_{t-1} \end{aligned}$$

where $O_t = \mathrm{diag}(P(e_t \mid X_t))$ is the observation matrix. By substitution,

$$O_1 = O_2 = O_4 = \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.2 \end{pmatrix} \quad \text{and} \quad O_3 = \begin{pmatrix} 0.2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.8 \end{pmatrix}.$$

Then, to calculate f_4, we first need f_1, f_2 and f_3.

$$f_1 = \alpha_1\, O_1\, T^T\, f_0 = \alpha_1 \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.2 \end{pmatrix} \begin{pmatrix} \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = \alpha_1 \begin{pmatrix} \frac{4}{15} \\ \frac{1}{6} \\ \frac{1}{15} \end{pmatrix} = \begin{pmatrix} \frac{8}{15} \\ \frac{1}{3} \\ \frac{2}{15} \end{pmatrix}$$

$$f_2 = \alpha_2\, O_2\, T^T\, f_1 = \dots$$

$$f_3 = \alpha_3\, O_3\, T^T\, f_2 = \dots$$

$$f_4 = \alpha_4\, O_4\, T^T\, f_3 = \dots \approx \begin{pmatrix} 0.472 & 0.374 & 0.154 \end{pmatrix}^T$$

(c) What are the probabilities of the coin picked first given the sequence of results? And of the first coin tossed?

Solution. We are asked to calculate the distribution $P(X_k \mid e_{1:t})$ for $t = 4$ and $k = 0$. As $k < t$ this corresponds to *smoothing* our belief of the past. We have

$$\begin{aligned} P(X_k \mid e_{1:t}) &= \alpha P(X_k, e_{k+1:t} \mid e_{1:k}) \\ &= \alpha P(e_{k+1:t} \mid X_k, e_{1:k}) P(X_k \mid e_{1:k}) \\ &= \alpha P(e_{k+1:t} \mid X_k) P(X_k \mid e_{1:k}) \end{aligned}$$

As we already know $f_k = P(X_k \mid e_{1:k})$, we only need to compute $b_k = P(e_{k+1:t} \mid X_k)$. We have

$$\begin{aligned} P(e_{k+1:t} \mid X_k) &= \sum_{x_{k+1}} P(x_{k+1} \mid X_k) P(e_{k+1:t} \mid X_k, x_{k+1}) \\ &= \sum_{x_{k+1}} P(x_{k+1} \mid X_k) P(e_{k+1} \mid x_{k+1}) P(e_{k+2:t} \mid x_{k+1}) \\ \Leftrightarrow \qquad b_k &= T\, O_{k+1}\, b_{k+1} \end{aligned}$$

where $b_t = b_4 = P(\textit{anything} \mid X_4) = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$. Therefore, to calculate b_0, we first need b_3, b_2 and b_1.

$$b_3 = T\, O_4\, b_4 = \begin{pmatrix} \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.65 \\ 0.5 \\ 0.35 \end{pmatrix}$$

$$b_2 = T\, O_3\, b_3 = \dots$$

$$b_1 = T\, O_2\, b_2 = \dots$$

$$b_0 = T\, O_1\, b_1 = \dots \approx \begin{pmatrix} 0.076 & 0.055 & 0.035 \end{pmatrix}^T$$

Eventually,

$$P(X_0 \mid e_{1:4}) = \alpha b_0 \times f_0 \approx \begin{pmatrix} 0.457 & 0.331 & 0.212 \end{pmatrix}^T$$
$$P(X_1 \mid e_{1:4}) = \alpha b_1 \times f_1 \approx \begin{pmatrix} 0.580 & 0.329 & 0.091 \end{pmatrix}^T$$

(d) What is the most likely sequence of tossed coins?

Solution. The most likely sequence $x^*_{1:t}$ given the sequence of evidences $e_{1:t}$ is the one that satisfies

$$x^*_{1:t} = \arg\max_{x_{1:t}} P(x_{1:t} \mid e_{1:t})$$

or, equivalently,

$$\begin{aligned}
x^*_k &= \arg\max_{x_k} \left[\max_{x_{1:k-1}} P(x_{1:k}, x^*_{k+1:t} \mid e_{1:t}) \right] \\
&= \arg\max_{x_k} \left[\max_{x_{1:k-1}} \alpha P(x_{1:k}, x^*_{k+1:t}, e_{k+1:t} \mid e_{1:k}) \right] \\
&= \arg\max_{x_k} \left[\max_{x_{1:k-1}} \alpha P(e_{k+1:t} \mid x_{1:k}, x^*_{k+1:t}, e_{1:k}) P(x^*_{k+1:t} \mid x_{1:k}, e_{1:k}) P(x_{1:k} \mid e_{1:k}) \right] \\
&= \arg\max_{x_k} \left[\max_{x_{1:k-1}} \underbrace{\alpha P(e_{k+1:t} \mid x^*_{k+1:t}) P(x^*_{k+2:t} \mid x^*_{k+1})}_{\text{constant}} P(x^*_{k+1} \mid x_k) P(x_{1:k} \mid e_{1:k}) \right] \\
&= \arg\max_{x_k} P(x^*_{k+1} \mid x_k) \left[\max_{x_{1:k-1}} P(x_{1:k} \mid e_{1:k}) \right] \\
&= \arg\max_{x_k} P(x^*_{k+1} \mid x_k)\, m_k(x_k)
\end{aligned}$$

where

$$\begin{aligned}
m_k &= \max_{x_{1:k-1}} P(x_{1:k-1}, X_k \mid e_{1:k}) \\
&= \max_{x_{1:k-1}} \alpha P(x_{1:k-1}, X_k, e_k \mid e_{1:k-1}) \\
&= \max_{x_{1:k-1}} \alpha P(e_t \mid x_{1:k-1}, X_t, e_{1:k-1}) P(X_k \mid x_{1:k-1}, e_{1:k-1}) P(x_{1:k-1} \mid e_{1:k-1}) \\
&= \max_{x_{1:k-1}} \alpha P(e_k \mid X_k) P(X_k \mid x_{k-1}) P(x_{1:k-1} \mid e_{1:k-1}) \\
&= \alpha P(e_k \mid X_k) \max_{x_{k-1}} P(X_k \mid x_{k-1}) \max_{x_{1:k-2}} P(x_{1:k-1} \mid e_{1:k-1}) \\
&= \alpha P(e_k \mid X_k) \max_{x_{k-1}} P(X_k \mid x_{k-1})\, m_{k-1}(x_{k-1})
\end{aligned}$$

and $m_1 = P(X_1 \mid e_1) = f_1$. Therefore, we can iteratively build the vectors $m_1, m_2, \ldots, m_t$ to find the most likely last state

$$x^*_t = \arg\max_{x_t} m_t(x_t)$$

and, then, the most likely sequence with

$$x^*_k = \arg\max_{x_k} P(x^*_{k+1} \mid x_k)\, m_k(x_k).$$

Exercise 4 Ball on a spring

The "Foire de Liège" has a new attraction called "Super Spring Ultra Pro Max XXL" which consists in a ball attached to a spring on a platform. The participants take seat in the ball locked at some position. The ball is then released and pulled by the spring which makes it oscillate back and forth. After a few seconds, the ball is stopped by magnetic brakes. You notice that the final position of the ball is different each time. As the ball is opaque, you wonder if it is possible for the participants to guess where the ball stopped, given their perception of acceleration.

You more or less remember your Newtonian mechanics class and model the movement of the ball as a series of transitions

$$\begin{aligned} p_t &= p_{t-1} + \Delta t\, \dot{p}_{t-1} + \frac{1}{2}\Delta t^2\, \ddot{p}_{t-1} \\ \dot{p}_t &= \dot{p}_{t-1} + \Delta t\, \ddot{p}_{t-1} \\ \ddot{p}_t &= \rho w_{t-1} - \kappa\, p_{t-1} - \eta\, \dot{p}_{t-1} \\ w_t \mid w_{t-1} &\sim \mathcal{N}(\alpha w_{t-1}, \sigma_w^2) \end{aligned}$$

where p_t, $\dot{p}_t$ and $\ddot{p}_t$ are respectively the position, velocity and acceleration of the ball at timestep t, w_t is the wind at timestep t, Δt is the time elapsed between $t-1$ and t, ρ is the thrust coefficient of the wind, κ is the stiffness of the spring, η is the friction coefficient of the platform and α is the persistence of the wind. You estimate that the ball starts $10.0 \pm 0.5\,\mathrm{m}$ at the left of the spring with a negligible speed $0.0 \pm 0.1\,\mathrm{m\,s^{-1}}$ and acceleration $0.0 \pm 0.1\,\mathrm{m\,s^{-2}}$. You assume that the wind has a stationary distribution. You assume that the human perception of acceleration follows an unbiased Gaussian distribution. You also assume that a good model to describe the uncertainty in your initial guess of the ball's state is a Gaussian model, with the standard deviation given by your estimated bounds.

(a) You wish to predict the state of the ball given the perceptions of a participant. Define the components of a Kalman filter in this context.

Solution. The Kalman filter is a special case of the continuous Bayes filter, which assumes

- a Gaussian prior
$$p(x_0) = \mathcal{N}(x_0 \mid \mu_0, \Sigma_0),$$
- a linear Gaussian transition model
$$p(x_t \mid x_{t-1}) = \mathcal{N}(x_t \mid F x_{t-1} + u, \Sigma_x)$$
- and a linear Gaussian sensor model
$$p(e_t \mid x_t) = \mathcal{N}(e_t \mid H x_t + v, \Sigma_e).$$

In a multivariate Gaussian distribution $x = \begin{pmatrix} x_1 & x_2 & \dots & x_n \end{pmatrix}^T \sim \mathcal{N}(\mu, \Sigma)$, the first argument is the *mean vector* and the second is the *covariance matrix*. The mean vector is the vector of the variable means, *i.e.* $\mu_i = \mathbb{E}[x_i]$. An element Σ_{ij} of the covariance matrix is the covariance between the variables x_i and x_j, *i.e.* $\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)]$. If x_i is independent from x_j, their covariance is null by definition, *i.e.* $\Sigma_{ij} = 0$. Interestingly, the diagonal elements Σ_{ii} are the variable variances $\mathbb{V}[x_i] = \mathbb{E}[(x_i - \mu_i)^2]$.

As a Kalman filter always relies on the same assumptions, the methodology outlined below can be applied directly, regardless of the problem under consideration:

1. identify the state variables;
2. identify the linear system, and derive the transition model and the sensor model;
3. identify the prior $p(x_0)$;
4. deduce $p(x_t \mid e_{1:t})$ using the Gaussian identities (point 2 of this exercise).

a) Identify the states variables

The state X_t comprises all the dynamic variables (*i.e.* the variables that evolve over time) of the system.

In our case, the state X_t is a 4-dimensional vector $\begin{pmatrix} p_t & \dot{p}_t & \ddot{p}_t & w_t \end{pmatrix}^T$.

b) Identify the linear system, and derive the transition model and the sensor model

The Kalman filter assumes a linear system with Gaussian noise. A canonical representation of such a system is as follows:

$$\begin{aligned} X_t &= FX_{t-1} + u + \epsilon \\ E_t &= HX_t + v + \gamma \end{aligned}$$

where $\epsilon \sim \mathcal{N}(0, \Sigma_x)$ represents the Gaussian uncertainty in the transition model, and $\gamma \sim \mathcal{N}(0, \Sigma_e)$ represents the Gaussian sensor noise.

By recalling the fundamental properties of Gaussian distributions, the distributions of the transition model and the sensor model can be readily derived from the canonical representation.

For $X_t \mid X_{t-1}$:

$$\begin{aligned} X_t \mid X_{t-1} &\sim \mathcal{N}(\cdot) \quad \text{(Sum of Gaussian r.v. with scalars)} \\ \mathbb{E}[X_t \mid X_{t-1}] &= \mathbb{E}[FX_{t-1} + u + \epsilon \mid X_{t-1}] \\ &= \mathbb{E}[FX_{t-1} \mid X_{t-1}] + \mathbb{E}[u] + \mathbb{E}[\epsilon] \\ &= FX_{t-1} + u \\ \mathbb{V}[X_t \mid X_{t-1}] &= \mathbb{V}[FX_{t-1} + u + \epsilon \mid X_{t-1}] \\ &= \mathbb{V}[FX_{t-1} \mid X_{t-1}] + \mathbb{V}[u] + \mathbb{V}[\epsilon] \\ &= \Sigma_x \\ \Rightarrow p(x_t \mid x_{t_1}) &= \mathcal{N}\left(x_t \mid Fx_{t-1} + u, \Sigma_x\right) \end{aligned}$$

For $E_t \mid X_t$:

$$\begin{aligned} E_t \mid X_t &\sim \mathcal{N}(\cdot) \quad \text{(Sum of Gaussian r.v. with scalars)} \\ \mathbb{E}[E_t \mid X_t] &= \mathbb{E}[HX_t + v + \gamma \mid X_t] \\ &= \mathbb{E}[HX_t \mid X_t] + \mathbb{E}[v] + \mathbb{E}[\gamma] \\ &= HX_t + v \\ \mathbb{V}[E_t \mid X_t] &= \mathbb{V}[HX_t + v + \gamma \mid X_t] \\ &= \mathbb{V}[HX_t \mid X_t] + \mathbb{V}[v] + \mathbb{V}[\gamma] \\ &= \Sigma_e \\ \Rightarrow p(e_t \mid x_t) &= \mathcal{N}\left(e_t \mid Hx_t + v, \Sigma_e\right) \end{aligned}$$

In our case, the linear system that must be identified is as follows:

$$\begin{pmatrix} p_t \\ \dot{p}_t \\ \ddot{p}_t \\ w_t \end{pmatrix} = \begin{pmatrix} F_{11} & F_{12} & F_{13} & F_{14} \\ F_{21} & F_{22} & F_{23} & F_{24} \\ F_{31} & F_{32} & F_{33} & F_{34} \\ F_{41} & F_{42} & F_{43} & F_{44} \end{pmatrix} \begin{pmatrix} p_{t-1} \\ \dot{p}_{t-1} \\ \ddot{p}_{t-1} \\ w_{t-1} \end{pmatrix} + \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} + \epsilon$$

$$\begin{pmatrix} e_t \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} & H_{13} & H_{14} \end{pmatrix} \begin{pmatrix} p_t \\ \dot{p}_t \\ \ddot{p}_t \\ w_t \end{pmatrix} + \begin{pmatrix} v_1 \end{pmatrix} + \gamma$$

Although it is easy from the statement to identify the F and u factors for the first three states (p_t, $\dot{p}_t$, $\ddot{p}_t$), the F and u factors associated with w_t, *i.e.* $F_{4,:}$ and u_4, are difficult to identify because the transition model for w_t is not given in the canonical form.
By recalling the fundamental properties of Gaussian distributions, the canonical form can be easily recovered:

$$w_t \mid w_{t-1} \sim \mathcal{N}(\alpha w_{t-1}, \sigma_w^2) \quad \Leftrightarrow \quad w_t = \alpha w_{t-1} + 0 + \xi,\ \xi \sim \mathcal{N}(0, \sigma_w^2)$$

Convincing oneself of the equivalence between the two formulae above can be done quickly by inspecting the type of distribution, the mean, and the variance for w_t in the right-hand side expression, as was done for $X_t \mid X_{t-1}$ or $E_t \mid X_t$.
The matrices F and H can now be found by identification:

$$\begin{pmatrix} p_t \\ \dot{p}_t \\ \ddot{p}_t \\ w_t \end{pmatrix} = \begin{pmatrix} 1 & \Delta t & \frac{1}{2}\Delta t^2 & 0 \\ 0 & 1 & \Delta t & 0 \\ -\kappa & -\eta & 0 & \rho \\ 0 & 0 & 0 & \alpha \end{pmatrix} \begin{pmatrix} p_{t-1} \\ \dot{p}_{t-1} \\ \ddot{p}_{t-1} \\ w_{t-1} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \epsilon$$

$$\begin{pmatrix} e_t \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} p_t \\ \dot{p}_t \\ \ddot{p}_t \\ w_t \end{pmatrix} + \begin{pmatrix} 0 \end{pmatrix} + \gamma$$

It remains only to determine the uncertainty in the transition model and the sensor noise, *i.e.* to specify:

$$\epsilon \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \Sigma_{x,11} & \Sigma_{x,12} & \Sigma_{x,13} & \Sigma_{x,14} \\ \Sigma_{x,21} & \Sigma_{x,22} & \Sigma_{x,23} & \Sigma_{x,24} \\ \Sigma_{x,31} & \Sigma_{x,32} & \Sigma_{x,33} & \Sigma_{x,34} \\ \Sigma_{x,41} & \Sigma_{x,42} & \Sigma_{x,43} & \Sigma_{x,44} \end{pmatrix} \right)$$

$$\gamma \sim \mathcal{N}\left(\begin{pmatrix} 0 \end{pmatrix}, \begin{pmatrix} \Sigma_{e,11} \end{pmatrix} \right)$$

No uncertainty is associated with the transition model of the first three states $\begin{pmatrix} p_t & \dot{p}_t & \ddot{p}_t \end{pmatrix}$ and the uncertainty on the wind is σ_w^2. Therefore, $\Sigma_{x,44}$ is the only non-zero component of Σ_x and is equal to σ_w^2. Regarding the sensor model, the uncertainty is not given. Let us call it σ_e^2.
To sum up, the transition model is given by:

$$p(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t \mid \begin{pmatrix} 1 & \Delta t & \frac{1}{2}\Delta t^2 & 0 \\ 0 & 1 & \Delta t & 0 \\ -\kappa & -\eta & 0 & \rho \\ 0 & 0 & 0 & \alpha \end{pmatrix} x_{t-1} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_w^2 \end{pmatrix} \right)$$

The sensor model is given by:

$$p(e_t \mid x_t) = \mathcal{N}\left(e_t \mid \begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix} x_t + \begin{pmatrix} 0 \end{pmatrix}, \sigma_e^2 \right).$$

c) Identify the prior

$$p\left(\begin{pmatrix} p_0 \\ \dot{p}_0 \\ \ddot{p}_0 \\ w_0 \end{pmatrix} \right) = \mathcal{N}\left(\begin{pmatrix} p_0 \\ \dot{p}_0 \\ \ddot{p}_0 \\ w_0 \end{pmatrix} \mid \begin{pmatrix} \mu_{0,1} \\ \mu_{0,2} \\ \mu_{0,3} \\ \mu_{0,4} \end{pmatrix}, \begin{pmatrix} \Sigma_{0,11} & \Sigma_{0,12} & \Sigma_{0,13} & \Sigma_{0,14} \\ \Sigma_{0,21} & \Sigma_{0,22} & \Sigma_{0,23} & \Sigma_{0,24} \\ \Sigma_{0,31} & \Sigma_{0,32} & \Sigma_{0,33} & \Sigma_{0,34} \\ \Sigma_{0,41} & \Sigma_{0,42} & \Sigma_{0,43} & \Sigma_{0,44} \end{pmatrix} \right)$$

From the statement, the following distributions are straightforward:

$$\begin{aligned} p_0 &\sim \mathcal{N}(10, 0.5^2) \\ \dot{p}_0 &\sim \mathcal{N}(0, 0.1^2) \\ \ddot{p}_0 &\sim \mathcal{N}(0, 0.1^2) \end{aligned}$$

The stationarity of the wind implies that the variance and the mean are time invariant:

$$\begin{aligned} & \mathbb{V}[w_0] = \mathbb{V}[w_1] = \mathbb{V}[\alpha w_0 + \xi] = \alpha^2 \mathbb{V}[w_0] + \sigma_w^2 \\ \Rightarrow\ & \mathbb{V}[w_0] = \frac{\sigma_w^2}{1-\alpha^2} \\ & \mathbb{E}[w_0] = \mathbb{E}[w_1] = \mathbb{E}[\alpha w_0 + \xi] = \alpha \mathbb{E}[w_0] + 0 \\ \Rightarrow\ & \mathbb{E}[w_0] = 0 \quad \text{(Assuming } \alpha \neq 0\text{)} \\ \Rightarrow\ & w_0 \sim \mathcal{N}(0, \frac{\sigma_w^2}{1-\alpha^2}) \end{aligned}$$

As no links between the uncertainties of the initial states are mentioned in the statement, we can assume that the initial states are independent. Therefore, $\Sigma_{0,ij} = 0 \quad \forall i \neq j$. The prior is:

$$p\left(\begin{pmatrix} p_0 \\ \dot{p}_0 \\ \ddot{p}_0 \\ w_0 \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} p_0 \\ \dot{p}_0 \\ \ddot{p}_0 \\ w_0 \end{pmatrix} \mid \begin{pmatrix} 0.5^2 & 0 & 0 & 0 \\ 0 & 0.1^2 & 0 & 0 \\ 0 & 0 & 0.1^2 & 0 \\ 0 & 0 & 0 & \frac{\sigma_w^2}{1-\alpha^2} \end{pmatrix}\right)$$

(b) Express the distribution $p(x_t \mid e_{1:t})$ with respect to the components defined previously. Use the cheat sheet for Gaussian models.

Solution. This task corresponds exactly to filtering, *i.e.* inferring the distribution

$$\begin{aligned} p(x_t \mid e_{1:t}) &= \alpha\, p(e_t \mid x_t, e_{1:t-1})\, p(x_t \mid e_{1:t-1}) \\ &= \alpha\, p(e_t \mid x_t) \int p(x_t, x_{t-1} \mid e_{1:t-1})\, \mathrm{d}x_{t-1}. \end{aligned}$$

In the latter expression, the integral corresponds to the marginalization of the joint distribution

$$p(x_t, x_{t-1} \mid e_{1:t-1}) = p(x_t \mid x_{t-1})\, p(x_{t-1} \mid e_{1:t-1}).$$

As all distributions (previous belief, transition model and sensor model)[1] are linear Gaussian, the distribution $p(x_t \mid e_{1:t})$ is also linear Gaussian and its expression can be found thanks to the Gaussian cheat sheet.

Understanding all the mathematical steps done to recover the expression:

$$p(x_t \mid e_{1:t}) = \alpha\, p(e_t \mid x_t, e_{1:t-1}) \int p(x_t \mid, x_{t-1}) p(x_{t-1} \mid e_{1:t-1})\, \mathrm{d}x_{t-1}$$

allows to easily use the cheat sheet.

1. $p(x_t \mid e_{1:t}) = \alpha\, p(e_t \mid x_t, e_{1:t-1})\, p(x_t \mid e_{1:t-1})$
   The conditioning of x_t on e_t is reversed.
   The cheat sheet tells us that, from $p(e_t \mid x_t, e_{1:t-1})$[2] and $p(x_t \mid e_{1:t-1})$, whose analytical expressions are easily deductible, we can recover the expression of the joint distribution:

$$p\begin{pmatrix} x_t \mid e_{1:t-1} \\ e_t \mid e_{1:t-1} \end{pmatrix}.$$

[1] Since the prior $p(x_0)$ is Gaussian by assumption, all beliefs are Gaussian as well, by induction.
[2] Equivalent to $p(e_t \mid x_t)$ under the Markov assumption.

Here, x_t plays the role of "x" in the cheat sheet, e_t plays the role of "y", and all distributions are conditioned on $e_{1:t-1}$.
From the recovered joint, it is easy to reverse the conditioning with the cheat sheet:

$$\text{“}p(x \mid y)\text{”} \qquad \Rightarrow \qquad p(x_t \mid e_t, e_{1:t-1})$$

Although the analytical expression of $p(e_t \mid x_t)$ is known (it is the sensor model), some work is still required to derive the analytical expression of $p(x_t \mid e_{1:t-1})$.

2. $p(x_t \mid e_{1:t-1}) = \int p(x_t \mid x_{t-1}, e_{1:t-1}) p(x_{t-1} \mid e_{1:t-1}) dx_{t-1}$
The distribution $p(x_t, x_{t-1} \mid e_{1:t-1})$ is marginalised over $x_{t-1} \mid e_{1:t-1}$.
From $p(x_t \mid x_{t-1}, e_{1:t-1})$ (the sensor model)[3] and $p(x_{t-1} \mid e_{1:t-1})$ (the previous belief),we can recover the expression of the joint distribution:

$$p \begin{pmatrix} x_{t-1} \mid e_{1:t-1} \\ x_t \mid e_{1:t-1} \end{pmatrix}.$$

Here, x_{t-1} plays the role of "x" in the cheat sheet, x_t plays the role of "y", and all distributions are conditioned on $e_{1:t-1}$.
From the joint distribution, we can recover the analytical expression of $p(x_t \mid e_{1:t-1})$, which corresponds to "$p(y)$" in the cheat sheet.

3. Putting all together
Steb (b)

$$p\left(\begin{pmatrix} x_{t-1} \\ x_t \end{pmatrix} \middle| e_{1:t-1}\right) = \mathcal{N}\left(\begin{pmatrix} x_{t-1} \\ x_t \end{pmatrix} \middle| \begin{pmatrix} \mu_{t-1} \\ F\mu_{t-1} + u \end{pmatrix}, \begin{pmatrix} \Sigma_{t-1} & \Sigma_{t-1}F^T \\ F\Sigma_{t-1} & F\Sigma_{t-1}F^T + \Sigma_x \end{pmatrix}\right)$$

$$\Rightarrow p(x_t \mid e_{1:t-1}) = \mathcal{N}\big(x_t \mid \underbrace{F\mu_{t-1} + u}_{\mu_*}, \underbrace{F\Sigma_{t-1}F^T + \Sigma_x}_{\Sigma_*}\big).$$

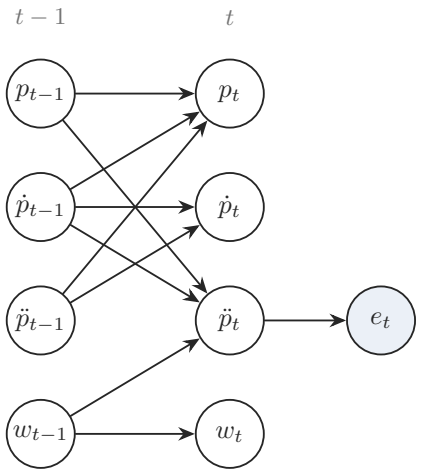
step (a)

$$p\left(\begin{pmatrix} x_t \\ e_t \end{pmatrix} \middle| e_{1:t-1}\right) = \mathcal{N}\left(\begin{pmatrix} x_t \\ e_t \end{pmatrix} \middle| \begin{pmatrix} \mu_* \\ H\mu_* + v \end{pmatrix}, \begin{pmatrix} \Sigma_* & \Sigma_* H^T \\ H\Sigma_* & H\Sigma_* H^T + \Sigma_e \end{pmatrix}\right).$$

$$\Rightarrow p(x_t \mid e_{1:t}) = \mathcal{N}\big(x_t \mid \underbrace{\mu_* + K(e_t - H\mu_* - v)}_{\mu_t}, \underbrace{\Sigma_* - KH\Sigma_*}_{\Sigma_t}\big)$$

where $K = \Sigma_* H^T (H\Sigma_* H^T + \Sigma_e)^{-1}$ is the Kalman gain matrix. The matrices F, H, Σ_x, Σ_y, u, v are the same as those defined in the previous point.

(c) Represent the transition and sensor models as a dynamic Bayesian network.
Solution.



[3] Equivalent to $p(x_t \mid x_{t-1})$ under the Markov assumption.

Exercise 5 Robot state estimation (Exam, January 2024)
Let us consider a continuous first-order Markov process describing the motion of a robot in a one-dimensional world. On average, the robot moves forward by a distance b at each timestep. The robot's position x_t at time t is not directly observable, but can be estimated using a sensor that measures the distance between the robot and the sensor. Assuming,

- the sensor is fixed at position 0,
- a Gaussian prior $\mathcal{N}(x_0 \mid \mu_0, \sigma_0^2)$ on the initial position of the robot,
- a Gaussian transition model $\mathcal{N}(x_{t+1} \mid x_t + b, \sigma_x^2)$,
- a Gaussian observation model $\mathcal{N}(e_t \mid x_t, \sigma_e^2)$,

derive the update equations of the Bayes filter. More specifically, since the belief distribution is Gaussian, derive the update equations of its mean μ_{t+1} and variance σ_{t+1}^2.

Do the update equations make sense in the following limiting cases? Use your physical intuition.

- The sensor is extremely noisy.
- The sensor is perfect.
- The transition model is extremely noisy.
- The transition model is perfect.

Solution.

1. Identify the states variables
   x_t, the robot's position.
2. Identify the linear system, and derive the transition model and the sensor model

$$\begin{aligned} X_t &= FX_{t-1} + u + \epsilon \\ E_t &= HX_t + v + \gamma \\ \epsilon \sim \mathcal{N}(0, \Sigma_x) \quad \gamma &\sim \mathcal{N}(0, \Sigma_e) \end{aligned}$$

   with $F = 1$, $u = b$, $\Sigma_x = \sigma_x^2$, $H = 1$, $v = 0$, $\Sigma_e = \sigma_e^2$.
3. Identify the prior
   $p(x_0) = \mathcal{N}(x_0 \mid \mu_0, \sigma_0^2)$
4. Putting all together
   Marginalization step

$$\begin{cases} p(\mathbf{x}) & = \mathcal{N}(\mathbf{m}, \mathbf{P}) \\ p(\mathbf{y} \mid \mathbf{x}) & = \mathcal{N}(\mathbf{Hx} + \mathbf{u}, \mathbf{R}) \\ p\begin{pmatrix}\mathbf{x} \\ \mathbf{y}\end{pmatrix} & = \mathcal{N}\left(\begin{pmatrix}\mathbf{m} \\ \mathbf{Hm} + \mathbf{u}\end{pmatrix}, \begin{pmatrix}\mathbf{P} & \mathbf{PH^T} \\ \mathbf{HP} & \mathbf{HPH^T} + \mathbf{R}\end{pmatrix}\right) \\ & = \mathcal{N}\left(\begin{pmatrix}\mathbf{a} \\ \mathbf{b}\end{pmatrix}, \begin{pmatrix}\mathbf{A} & \mathbf{C} \\ \mathbf{C^T} & \mathbf{B}\end{pmatrix}\right) \\ p(\mathbf{y}) & = \mathcal{N}(\mathbf{b}, \mathbf{B}) \end{cases}$$

$$\Leftrightarrow$$

$$\begin{cases} p(x_{t-1} \mid e_{1:t-1}) & = \mathcal{N}(\mu_{t-1}, \sigma_{t-1}^2) \\ p(x_t \mid x_{t-1}, e_{1:t-1}) & = \mathcal{N}(1 \cdot x_{t-1} + b, \sigma_x^2) \\ p\left(\begin{pmatrix} x_{t-1} \\ x_t \end{pmatrix} \middle| e_{1:t-1}\right) & = \mathcal{N}\left(\begin{pmatrix} \mu_{t-1} \\ 1 \cdot \mu_{t-1} + b \end{pmatrix}, \begin{pmatrix} \sigma_{t-1}^2 & \sigma_{t-1}^2 \cdot 1 \\ 1 \cdot \sigma_{t-1}^2 & 1 \cdot \sigma_{t-1}^2 \cdot 1 + \sigma_x^2 \end{pmatrix}\right) \\ p(x_t \mid e_{1:t-1}) & = \mathcal{N}(\mu_{t-1} + b, \sigma_{t-1}^2 + \sigma_x^2) \end{cases}$$

Reverse the conditioning

$$\begin{cases} p(\mathbf{x}) & = \mathcal{N}(\mathbf{m}, \mathbf{P}) \\ p(\mathbf{y} \mid \mathbf{x}) & = \mathcal{N}(\mathbf{Hx} + \mathbf{u}, \mathbf{R}) \\ p\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} & = \mathcal{N}\left(\begin{pmatrix} \mathbf{m} \\ \mathbf{Hm} + \mathbf{u} \end{pmatrix}, \begin{pmatrix} \mathbf{P} & \mathbf{PH^T} \\ \mathbf{HP} & \mathbf{HPH^T} + \mathbf{R} \end{pmatrix}\right) \\ & = \mathcal{N}\left(\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C^T} & \mathbf{B} \end{pmatrix}\right) \\ p(\mathbf{x} \mid \mathbf{y}) & = \mathcal{N}(\mathbf{a} + \mathbf{CB^{-1}}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{CB^{-1}C^T}) \end{cases}$$

$$\Leftrightarrow$$

$$\begin{cases} p(x_t \mid e_{1:t-1}) & = \mathcal{N}(\mu_{t-1} + b, \sigma_{t-1}^2 + \sigma_x^2) \\ p(e_t \mid x_t, e_{1:t-1}) & = \mathcal{N}(1 \cdot x_t + 0, \sigma_e^2) \\ p\left(\begin{pmatrix} x_t \\ e_t \end{pmatrix} \middle| e_{1:t-1}\right) & = \mathcal{N}\left(\begin{pmatrix} \mu_{t-1} + b \\ 1 \cdot (\mu_{t-1} + b) + 0 \end{pmatrix}, \begin{pmatrix} \sigma_{t-1}^2 + \sigma_x^2 & (\sigma_{t-1}^2 + \sigma_x^2) \cdot 1 \\ 1 \cdot (\sigma_{t-1}^2 + \sigma_x^2) & 1 \cdot (\sigma_{t-1}^2 + \sigma_x^2) \cdot 1 + \sigma_e^2 \end{pmatrix}\right) \\ p(x_t \mid e_{1:t}) & = \mathcal{N}\Big(\mu_{t-1} + b + \frac{\sigma_{t-1}^2 + \sigma_x^2}{\sigma_{t-1}^2 + \sigma_x^2 + \sigma_e^2}(e_t - (\mu_{t-1} + b)), \\ & \qquad \sigma_{t-1}^2 + \sigma_x^2 - \frac{(\sigma_{t-1}^2 + \sigma_x^2)^2}{\sigma_{t-1}^2 + \sigma_x^2 + \sigma_e^2}\Big) \end{cases}$$

The update equations of the mean μ_{t+1} and the variance σ_{t+1}^2 are:

$$\begin{aligned} \mu_{t+1} &= \mu_t + b + \frac{\sigma_t^2 + \sigma_x^2}{\sigma_t^2 + \sigma_x^2 + \sigma_e^2}(e_{t+1} - (\mu_t + b)) \\ \sigma_{t+1}^2 &= (\sigma_t^2 + \sigma_x^2)(1 - \frac{\sigma_t^2 + \sigma_x^2}{\sigma_t^2 + \sigma_x^2 + \sigma_e^2}) \\ &= \frac{(\sigma_t^2 + \sigma_x^2)\sigma_e^2}{\sigma_t^2 + \sigma_x^2 + \sigma_e^2} \end{aligned}$$

Note: Although the systematic methodology is cumbersome for this simple exercise, it allows more complex problems to be solved easily (*cf.* previous exercise). An experimented student can go straight to the point!

- The sensor is extremely noisy.
  With $\sigma_e^2 \to +\infty$, the update equations become:

$$\begin{aligned} \mu_{t+1} &= \mu_t + b \\ \sigma_{t+1}^2 &= (\sigma_t^2 + \sigma_x^2) \end{aligned}$$

  As expected, the optimal strategy is to completely ignore the observations and rely solely on the initial belief and the transition model. The uncertainty is indeed the combination of the uncertainty in the transition model and the uncertainty of the previous state, which ultimately originates from the uncertainty in the initial belief.

- The sensor is perfect.
  With $\sigma_e^2 \to 0$, the update equations become:

$$\mu_{t+1} = e_{t+1}$$
$$\sigma_{t+1}^2 = 0$$

  As expected, the optimal strategy is to rely solely on the observations. Consequently, no uncertainty arises when estimating the robot's position.

- The transition model is extremely noisy.
  With $\sigma_x^2 \to +\infty$, the update equations become:

$$\begin{aligned}\mu_{t+1} &= e_{t+1} \\ \sigma_{t+1}^2 &= \frac{(\sigma_x^2)\sigma_e^2}{\sigma_x^2} \\ &= \sigma_e^2\end{aligned}$$

  As expected, the optimal strategy is to rely solely on the observations, since nothing reliable can be obtained after applying the transition model. Consequently, the uncertainty in the estimation is that of the sensor model.

- The transition model is perfect.
  With $\sigma_x^2 \to 0$, the update equations become:

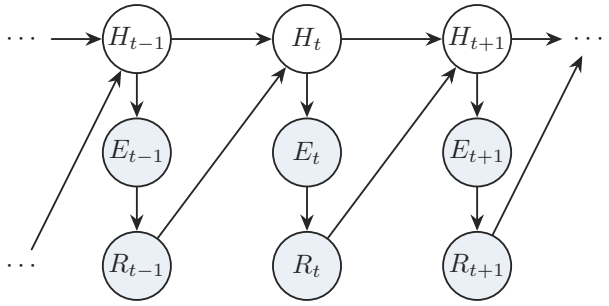
$$\begin{aligned}\mu_{t+1} &= \mu_t + b + \frac{\sigma_t^2}{\sigma_t^2 + \sigma_e^2}(e_{t+1} - (\mu_t + b)) \\ \sigma_{t+1}^2 &= (\sigma_t^2 + \sigma_x^2)(1 - \frac{\sigma_t^2 + \sigma_x^2}{\sigma_t^2 + \sigma_x^2 + \sigma_e^2}) \\ &= \frac{\sigma_t^2 \sigma_e^2}{\sigma_t^2 + \sigma_e^2}\end{aligned}$$

  Although it may seem surprising to still observe the effect of the observations on the estimation, the result is actually perfectly sensible when recalling that an uncertainty σ_0^2 is associated with the first estimate. If the initial position of the robot is known ($\sigma_0^2 \to 0$), then the update equations with the perfect transition model become:

$$\mu_{t+1} = \mu_t + b$$
$$\sigma_{t+1}^2 = 0$$

  As expected, the optimal strategy is to rely solely on the initial position and the transition model and no uncertainty is associated with the estimation.

Exercise 6 Human-robot interaction (CS188, Fall 2017)
In the near future, autonomous robots will live among us. Therefore, the robots need to know how to act appropriately in the presence of humans. In this question, we explore a simplified model of this interaction. We assume that we can observe the robot's actions at time t, R_t, and an evidence observation, E_t, directly caused by the (hidden) human state, H_t. Robot actions from the current time-step affect the human state in the next-time step, as illustrated in the Bayesian network given below.



Assuming discrete random variables and given the probability tables $P(H_0)$, $P(E_t \mid H_t)$, $P(R_t \mid E_t)$ and $P(H_{t+1} \mid H_t, R_t)$, your goal is to derive a procedure to maintain a belief $P(H_t \mid e_{1:t}, r_{1:t})$ about the state of the human at time t.

(a) Derive an update equation for incorporating a pair of observations (e_t, r_t) to a given belief state $P(H_t)$.

Solution.

$$\begin{aligned} P(H_t \mid e_t, r_t) &= \frac{P(e_t, r_t \mid H_t)P(H_t)}{P(e_t, r_t)} \\ &= \alpha P(r_t \mid e_t)P(e_t \mid H_t)P(H_t) \\ &= \alpha P(e_t \mid H_t)P(H_t) \end{aligned}$$

(b) Derive an update equation for predicting the future state H_{t+1} of the human at time $t+1$ given a belief state $P(H_t \mid e_{1:t}, r_{1:t})$.

Solution.

$$\begin{aligned} P(H_{t+1} \mid e_{1:t}, r_{1:t}) &= \sum_{h_t} P(H_{t+1}, h_t \mid e_{1:t}, r_{1:t}) \\ &= \sum_{h_t} P(H_{t+1} \mid h_t, r_t)P(h_t \mid e_{1:t}, r_{1:t}) \end{aligned}$$

(c) Combine both equations to derive a recursive update equation of the belief state $P(H_t \mid e_{1:t}, r_{1:t})$, as observations are collected and time passes.

Solution.

$$\begin{aligned} P(H_{t+1} \mid e_{1:t+1}, r_{1:t+1}) &= \alpha P(e_{t+1} \mid H_{t+1})P(H_{t+1} \mid e_{1:t}, r_{1:t}) \\ &= \alpha P(e_{t+1} \mid H_{t+1}) \sum_{h_t} P(H_{t+1} \mid h_t, r_t)P(h_t \mid e_{1:t}, r_{1:t}) \end{aligned}$$

(d) Let us now assume that all variables are continuous. Discuss how you would compute or approximate the belief state on a computer.

Solution. If all variables are continuous, unless the prior, transition model and sensor model are strictly constrained (*e.g.* Gaussian), it is not possible to express and update the belief states in closed form. Instead, we can use approximate filtering algorithms, such as extended Kalman filters or particle filters.

Exercise 7 **Server cooling** (Exam, August 2019) (AIMA ch. 14)
A monitoring program wants to know whether a server is cooled enough. Each day, the program checks whether the server raised a temperature alert, and whether it throttled its processors. The program makes the following hypotheses:

- The prior probability that the cooling is enough, with no observations, is 0.7.
- The probability that the cooling is enough on day t is 0.8 given that it was enough on the previous day, and 0.3 otherwise.
- The probability of a temperature alert is 0.2 if the cooling is enough, and 0.7 otherwise.
- The probability of throttling is 0.1 if the cooling is enough, and 0.3 otherwise.

(a) Formulate the environment and hypotheses as a dynamic Bayesian network that the program could use to detect insufficient cooling from a sequence of observations. Provide the associated probability tables.

(b) Reformulate the dynamic Bayesian network as a hidden Markov model that has only a single observation variable. Give the complete probability tables for the model.

(c) For the sequence $e_{1:3}$ of observations "no alert, no throttling", "alert, no throttling" and "alert, throttling", calculate the distributions $P(Cooled_t \mid e_{1:t})$ and $P(Cooled_t \mid e_{1:3})$ for $t \in \{1, 2, 3\}$.

Quiz

1. The normal distribution $\mathcal{N}(\mu, \sigma^2)$ is described by the density function
   - ○ $p(x) = \frac{1}{\sigma} \exp\left(-(z + \exp(-z))\right)$, with $z = \frac{x-\mu}{\sigma}$.
   - ○ $p(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x-\mu|}{\sigma}\right)$.
   - ● $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
   - ○ $p(x) = \dfrac{1}{\pi\sigma\left[1 + \left(\frac{x-\mu}{\sigma}\right)^2\right]}$.
2. The Kalman filter requires the specification of ...
   - ○ a prior $\mathbf{P}(\mathbf{X}_0)$ and a transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$.
   - ○ a transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$ and an observation model $\mathbf{P}(\mathbf{E}_t \mid \mathbf{x}_t)$.
   - ● a prior $\mathbf{P}(\mathbf{X}_0)$, a transition model $P(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$, and an observation model $P(\mathbf{E}_t \mid \mathbf{x}_t)$.
   - ○ a prior $\mathbf{P}(\mathbf{X}_0)$, a prior $\mathbf{P}(\mathbf{E}_0)$, a transition model $P(\mathbf{X}_{t+1} \mid \mathbf{x}_t)$, and an observation model $P(\mathbf{E}_t \mid \mathbf{x}_t)$.
3. In filtering, the belief state is updated at each timestep using ...
   - ○ the current belief and the transition model.
   - ○ the current belief, the observation model and the new evidence.
   - ○ the transition model, the observation model and the new evidence.
   - ● the current belief, the transition model, the observation model and the new evidence.

4. Let us assume a Markov process with hidden state variables $\mathbf{X}_t$, evidence variables $\mathbf{E}_t$, transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{X}_t)$ and sensor model $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$. Which of the following is true?

   ◯ The Bayes filter estimates $\mathbf{P}(\mathbf{X}_{1:t} \mid \mathbf{e}_{1:t})$.

   ◯ Smoothing consists in computing $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1})$ by pushing the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ forward through the transition model and then updating this new belief state with the evidence $\mathbf{e}_{t+1}$.

   ◯ The Kalman filter computes $\mathbf{P}(\mathbf{X}_{t+k} \mid \mathbf{e}_{1:t})$ by pushing the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ forward k times through the transition model.

   ◉ The Viterbi algorithm computes the most likely sequence of state values $\mathbf{x}_{1:t}$ given the evidence $\mathbf{e}_{1:t}$.